

Lecture Set 4
DNSC 8328
INTRODUCTION TO BAYESIAN STATISTICS

Refik Soyer

The George Washington University
Department of Decision Sciences

ANALYSIS OF MULTINOMIAL EXPERIMENTS

- ▶ What if we have $k > 2$ categories in each trial?
 $\theta_i = \text{Prob}\{\text{outcome is of type } i\}$ and $\sum_i^k \theta_i = 1$, that is, we have mutually exclusive outcomes.

Consider n independent trials where θ_i 's are constant over trials $\Rightarrow$ multinomial experiment.

- ▶ We are interested in random variables:

Y_i = number of outcomes of type i such that $\sum_i^k Y_i = n$.

The joint probability distribution of Y_i 's is multinomial given by

$$\begin{aligned} \text{Prob}\{Y_1 = y_1, Y_2 = y_2, \dots, Y_k = y_k | \theta_1, \theta_2, \dots, \theta_k\} \\ = \frac{n!}{y_1! y_2! \dots y_k!} \theta_1^{y_1} \theta_2^{y_2} \dots \theta_k^{y_k}, \end{aligned}$$

where $\sum_i^k y_i = n$ and $\sum_i^k \theta_i = 1$.

PROPERTIES OF THE MULTINOMIAL DISTRIBUTION

- Binomial is the special case where $k = 2$.

All marginals are $(Y_i|n, \theta_i) \sim \text{Bin}(n, \theta_i)$ with

$$E(Y_i|\theta_i) = n\theta_i, V(Y_i|\theta_i) = n\theta_i(1 - \theta_i).$$

All pairs negatively correlated with

$$\text{Cov}(Y_i, Y_j|\theta_i, \theta_j) = -n\theta_i\theta_j,$$

and all conditionals are binomial given by

$$(Y_i|Y_j = y, \theta_i, \theta_j) \sim \text{Bin}(n - y, \frac{\theta_i}{1 - \theta_j}).$$

- Bayesian inference on $\theta_1, \theta_2, \dots, \theta_k$

What type of prior for $\theta_1, \theta_2, \dots, \theta_k$?

DIRICHLET DISTRIBUTION AS A PRIOR

- ▶ Note that $\sum_i^k \theta_i = 1 \Rightarrow \theta_i$'s will be negatively correlated.

It is possible to develop a conjugate Bayesian analysis for the multinomial distribution using a *Dirichlet* prior distribution.

- ▶ Dirichlet distribution is a multivariate extension of the Beta distribution with density.

$$p(\theta_1, \theta_2, \dots, \theta_k) \propto \prod_{i=1}^k \theta_i^{a_i-1}$$

where $\sum_{i=1}^k \theta_i = 1$, and $a_1, a_2, \dots, a_k$ are specified parameters ("prior sample sizes for each category"), and $a_i > 0$ for all i .

The constant of proportionality is a ratio of gamma functions, that is,

$$p(\theta_1, \theta_2, \dots, \theta_k) = \frac{\Gamma(a_1 + \dots + a_k)}{\Gamma(a_1)\Gamma(a_2)\dots\Gamma(a_k)} \prod_{i=1}^k \theta_i^{a_i-1}.$$

PROPERTIES OF THE DIRICHLET DISTRIBUTION

- ▶ We can show that all marginals are beta densities as $\theta_i \sim \text{Beta}(a_i, a_0 - a_i)$ for all i , where $a_0 = \sum_{i=1}^k a_i$, that is, $E(\theta_i) = a_i/a_0$.

All pairs are negatively correlated with

$$\text{Cov}(\theta_i, \theta_j) = -\frac{a_i a_j}{a_0^2(a_0 + 1)}.$$

- ▶ If we set all $a_i = 1 \Rightarrow$ equal density (uniform) for any vector $(\theta_1, \theta_2, \dots, \theta_k)$.

If we set $a_i = 0$ for all $i \Rightarrow$ improper joint proper density, but the resulting joint posterior density will be proper given that we observe at least one observation in each category.

- ▶ Posterior analysis: given data $(y_1, y_2, \dots, y_k, n)$, the posterior distribution is

$$p(\theta_1, \theta_2, \dots, \theta_k | y_1, \dots, y_k, n) \propto [\theta_1^{y_1} \theta_2^{y_2} \dots \theta_k^{y_k}] \prod_{i=1}^k \theta_i^{a_i-1}.$$

POSTERIOR DISTRIBUTION

- ▶ The above can be written as

$$p(\theta_1, \theta_2, \dots, \theta_k | y_1, \dots, y_k, n) \propto \prod_{i=1}^k \theta_i^{a_i + y_i - 1}.$$

Thus, the posterior distribution of $(\theta_1, \theta_2, \dots, \theta_k)$ is a Dirichlet distribution with parameters $a_i^* = a_i + y_i$, $i = 1, \dots, k$.

- ▶ As in the prior distribution, we can show that the marginals are beta distributions,

$$(\theta_i | y_1, \dots, y_k, n) \sim \text{Beta}(a_i^*, a_0 + n - a_i^*)$$

where $\sum_{i=1}^k a_i^* = a_0 + n$.

- ▶ We can show that

$$E(\theta_i | y_1, \dots, y_k, n) = \frac{a_i + y_i}{a_0 + n} = w \frac{a_i}{a_0} + (1 - w) \frac{y_i}{n},$$

where $w = a_0 / (a_0 + n)$.

PREDICTIVE DISTRIBUTIONS

- ▶ The prior predictive distribution is obtained as a Dirichlet-multinomial distribution given by

$$p(y_1, \dots, y_k) = \frac{\Gamma(a_0)n!}{\prod_{i=1}^k \Gamma(a_i) \prod_{i=1}^k y_i!} \frac{\prod_{i=1}^k \Gamma(a_i + y_i)}{\Gamma(a_0 + n)}$$

- ▶ How?

$$p(y_1, \dots, y_k) = \int p(y_1, \dots, y_k | \theta_1, \dots, \theta_k) p(\theta_1, \dots, \theta_k) d\theta_1, \dots, d\theta_k.$$

$$p(y_1, \dots, y_k) = \int \left[\frac{n!}{\prod_{i=1}^k y_i!} \prod_{i=1}^k \theta_i^{y_i} \right] \left[\frac{\Gamma(a_0)}{\prod_{i=1}^k \Gamma(a_i)} \prod_{i=1}^k \theta_i^{a_i-1} \right] d\theta_1, \dots, d\theta_k$$

$$p(y_1, \dots, y_k) = \frac{\Gamma(a_0)}{\prod_{i=1}^k \Gamma(a_i)} \frac{n!}{\prod_{i=1}^k y_i!} \int \left[\prod_{i=1}^k \theta_i^{a_i+y_i-1} \right] d\theta_1, \dots, d\theta_k.$$

The posterior predictive distribution can be obtained by replacing a_i with a_i^* .

EXAMPLE: 1988 PRE-ELECTION POLLING

- ▶ CBS News survey [Gelman et. al. (2003), pp.83] of 1447 adults in October 1988 resulted in

$y_1 = 727$ expressed support for George Bush

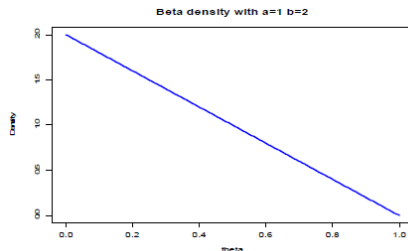
$y_2 = 583$ expressed support for Mike Dukakis

$y_3 = 137$ expressed support for others or no opinion.

Assuming a random sample is taken, $(y_1, y_2, y_3 | \theta_1, \theta_2, \theta_3)$ follows a multinomial distribution.

- ▶ We are interested in estimating posterior distribution of the difference $(\theta_1 - \theta_2)$.

Assume that $a_1 = a_2 = a_3 = 1$, then marginals for each $\theta_i \sim \text{Beta}(1, 2)$ as:



R CODE

Note that the joint posterior will be a Dirichlet distribution with parameters: $a_1^* = 728$, $a_2^* = 584$, and $a_3^* = 138$.

```
#Loading package LearnBayes
library(LearnBayes)
```

```
## Warning: package 'LearnBayes' was built under R version 4.0.3
```

```
# prior parameters and data
alpha=c(1,1,1)
y=c(727,583,137)
```

```
# posterior parameters and simulation
alphastar=alpha+y
theta=rdirichlet(5000,alphastar)
```

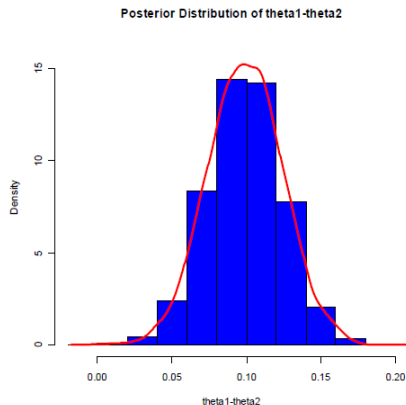
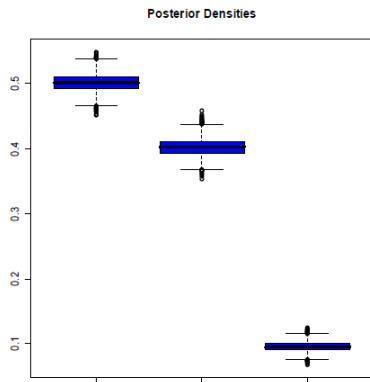
```
#Computing posterior probability of difference
DIF12=theta[,1]-theta[,2]
prob=mean(DIF12>0)
```

```
# Boxplot comparison and Posterior distribution of difference
```

```
par(mfrow=c(1,2))
boxplot(theta[,1],theta[,2],theta[,3],col="blue",main="Posterior Densities")

hist(DIF12,col="blue",xlab="theta1-theta2",main="Posterior Distribution of theta1-theta2",probability="true",ylim=c(0,16))
lines(density(DIF12),col="red",lwd=3)
```

POSTERIOR ANALYSIS



$$\text{Prob}\{\theta_1 > \theta_2 | y_1, y_2, y_3, n\} \approx 1.$$

BAYESIAN GOODNESS OF FIT TEST

- Consider the hypotheses

$$h_0 : (\theta_1 \theta_2 \dots \theta_k) = (\theta_1^* \theta_2^* \dots \theta_k^*)$$

$$h_1 : (\theta_1 \theta_2 \dots \theta_k) \neq (\theta_1^* \theta_2^* \dots \theta_k^*)$$

- Assume that $(\theta_1, \theta_2, \dots, \theta_k | h_1) \sim \text{Dir}(a_1, a_2, \dots, a_k)$ and we have observed data $D = \{(y_i, n_i); i = 1, \dots, k\}$, then we can obtain

$$p(D|h_1) = \frac{\Gamma(a_0)n!}{\prod_{i=1}^k \Gamma(a_i) \prod_{i=1}^k y_i!} \frac{\prod_{i=1}^k \Gamma(a_i + y_i)}{\Gamma(a_0 + n)},$$

and

$$p(D|h_0) = \frac{n!}{y_1! y_2! \dots y_k!} (\theta_1^*)^{y_1} (\theta_2^*)^{y_2} \dots (\theta_k^*)^{y_k}.$$

- Thus, the BF is given by

$$p(D|h_0)/p(D|h_1).$$

EXAMPLE

- ▶ Consider a multinomial experiment with $k = 3$ where we are interested in testing the hypotheses that

$$h_0 : \theta_1 = \theta_2 = \theta_3 = 1/3 \text{ versus } h_1 : \text{not}(\theta_1 = \theta_2 = \theta_3 = 1/3).$$

Under h_1 , assume that Dirichlet prior parameters are $a_i = 1, i = 1, \dots, 3$.

- ▶ We can show that with this prior

$$p(D|h_1) = \frac{\Gamma(a_0)n!}{\Gamma(a_0 + n)}.$$

If the observed data is $n = 6, y_1 = y_2 = y_3 = 2$, what is BF_{01} ?

$$p(D|h_0) = \frac{6!}{2!2!2!} (1/3)^6 = 10/81,$$

$$p(D|h_1) = \frac{\Gamma(a_0)n!}{\Gamma(a_0 + n)} = \frac{\Gamma(3)6!}{\Gamma(9)} = 1/28.$$

BAYES FACTORS

- We can obtain

$$BF_{01} = \frac{6!}{2!2!2!} (1/3)^6 (56/2) = 280/81 = 3.457.$$

- Note that if $n = 6, y_1 = 4, y_2 = 1, y_3 = 1$ then

$$BF_{01} = \frac{6!}{4!1!1!} (1/3)^6 (56/2) = 280/243 = 1.152.$$

- If we have $n = 6, y_1 = 6, y_2 = 0, y_3 = 0$ then

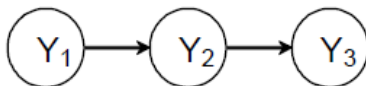
$$BF_{01} = \frac{6!}{6!0!0!} (1/3)^6 (56/2) = 28/729 = 0.038.$$

BAYES NETWORKS

- ▶ What is a Bayes network ?
 - ▶ Bayesian networks are graphical models describing probabilistic relationships (conditional independences) among a set of random quantities.
- ▶ Characteristics of Bayes networks:
 - ▶ They enable us model dependencies and conditional independencies,
 - ▶ They enable us to learn from data about the parameters of the associated probability distributions,
 - ▶ They can handle incomplete data,
 - ▶ They enable us to learn about the model structure from data (learning about "causal relationships"),
 - ▶ They can do all this adaptively as new data becomes available.

DIRECTED ACYCLIC GRAPHS (DAGs)

- ▶ A Bayesian network (Bayesian Belief Network-BBN) consists of:
- ▶ A set of random variables/quantities $Y = \{Y_1, \dots, Y_I\}$ and a Directed Acyclic Graph (DAG) defining a model M .
- ▶ A DAG is a graph with directed edges (arrows) between pairs of nodes (representing random quantities in Y) and with no directed cycles.
- ▶ The DAG represents (conditional) independence relationships between the variables implied by the given model M .



- ▶ The graph represents the conditional dependencies and enables us to factor the joint distribution of the variables by using the representation.

CONDITIONAL INDEPENDENCIES

- ▶ In the above DAG, we have Y_1 and Y_3 are conditionally independent given Y_2 denoted as

$$Y_1 \perp Y_3 | Y_2$$

- ▶ Thus, when we write the joint probability distribution of Y_i 's using the multiplication rule (chain rule), we have

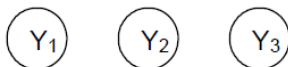
$$p(Y_1, Y_2, Y_3) = p(Y_1)p(Y_2|Y_1)p(Y_3|Y_1, Y_2) = p(Y_1)p(Y_2|Y_1)p(Y_3|Y_2)$$

due to the conditional independence implied by the DAG.

- ▶ Note that
 1. At node Y_1 : we have $p(Y_1) \Rightarrow Y_1$ is the only parent of Y_2 .
 2. At node Y_2 : we have $p(Y_2|Y_1) \Rightarrow Y_2$ is the only parent of Y_3 .
 3. At node Y_3 : we have $p(Y_3|Y_2) \Rightarrow Y_1$ has no parent.

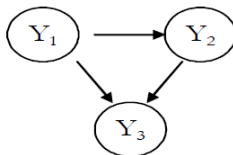
OTHER POSSIBLE MODEL STRUCTURES

- ▶ (Mutual) independence:



$\Rightarrow$ no connections; no node has a child or parent

- ▶ Complete dependence:

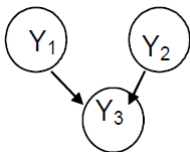


Here Y_2 has one parent, that is, Y_1 ; Y_3 has two parents Y_1 and Y_2 .

Thus, $p(Y_1, Y_2, Y_3) = p(Y_1)p(Y_2|Y_1)p(Y_3|Y_1, Y_2)$.

BASICS OF BAYES NETWORKS

- ▶ We have I random variables, $Y = \{Y_1, \dots, Y_I\}$ and a DAG for a given model M .
- ▶ Consider categorical variables, that is, each Y_i has C_i (mutually exclusive) categories for $i = 1, \dots, I$ and let y_{ik} represent a state of Y_i , $k = 1, \dots, C_i$.
- ▶ Define Π_i = parents of variable (that is, node) i and $\pi_i = (\pi_{i1}, \dots, \pi_{iQ_i})$ represent Q_i different configurations of parents of Y_i .
- ▶ Example: Consider the DAG where all the variables are binary, that is, $C_i = 2$ for all i .



- ▶ $\Pi_1 = \Pi_2 = \emptyset$ and $\Pi_3 = \{Y_1, Y_2\} \Rightarrow Q_3 = 4$ and
 $\pi_3 = \{\pi_{31} = (y_{11}, y_{21}), \pi_{32} = (y_{11}, y_{22}), \pi_{33} = (y_{12}, y_{21}), \pi_{34} = (y_{12}, y_{22})\}$

AN IMPORTANT PROPERTY OF DAG's

- ▶ In general, the DAG structure represents the assumption that a variable Y_i is independent of all its non-descendants given its parents Π_i .
- ▶ In obtaining the joint distribution of $\mathbf{y} = (y_1, \dots, y_I)$

$$p(\mathbf{y}) = \prod_{i=1}^I p(y_i | y_1, \dots, y_{i-1})$$

if we know that for every Y_i , where nodes are ordered in an ancestral order, there is $\Pi_i \subseteq \{Y_1, Y_2, \dots, Y_{i-1}\}$ such that given Π_i , Y_i is independent of all of its non-descendants, then we can write

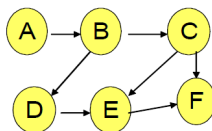
$$p(\mathbf{y}) = \prod_{i=1}^I p(y_i | y_1, \dots, y_{i-1}) = \prod_{i=1}^I p(y_i | \Pi_i)$$

Thus, in our case

$$p(y_1, y_2, y_3) = \prod_{i=1}^3 p(y_i | \Pi_i) = p(y_1)p(y_2)p(y_3 | y_1, y_2)$$

WHAT IS THE BIG DEAL ?

- Consider the following DAG structure



where $A, \dots, E, F$ are all binary variables, that is, $C_i = 2$ for all i .

- We have $\Pi_A = \emptyset$, $\Pi_B = \{A\}$, $\Pi_C = \{B\}$, $\Pi_D = \{B\}$, $\Pi_E = \{C, D\}$ and $\Pi_F = \{C, E\}$. Thus, using the general rule, we can write

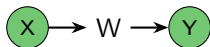
$$p(A, \dots, F) = p(A)p(B|A)p(C|B)p(D|B)p(E|C, D)p(F|C, E)$$

- Since we have binary variables to evaluate the joint for all combinations, we need the estimates for $1 + 2 + 2 + 2 + 4 + 4 = 15$ probabilities.
- If we do not have the structure we need $2^6 - 1 = 63$ probability estimates.

D-SEPARATION: INDEPENDENCE OFF THE GRAPH

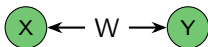
D-separation is a rule that decides, from the graph alone, whether two nodes X and Y are conditionally independent given a set Z .

Picture information trying to flow along a path between them; the path is *blocked* when it cannot pass through its middle node W .



Chain

blocked iff $W \in Z$



Fork

blocked iff $W \in Z$



Collider

blocked iff $W \notin Z$ and no descendant of W in Z

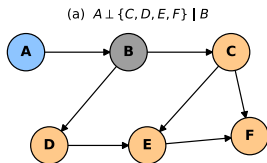
(Shaded node = observed, i.e. in the conditioning set Z .)

Definition

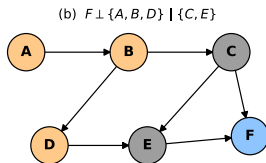
X and Y are *d-separated* by Z if every path between them is blocked.

Then $X \perp\!\!\!\perp Y \mid Z$.

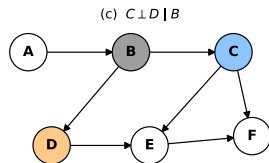
USING D-SEPARATION



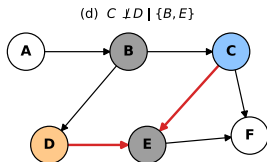
B is a chain node on every path from A



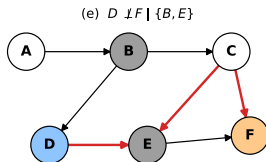
Parents of F = Markov blanket of F



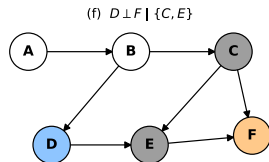
Fork at B blocked; colliders E, F closed



Conditioning on collider E opens $C \leftrightarrow E \leftrightarrow D$



Open path $D \rightarrow E \leftarrow C \rightarrow F$ (C not observed)

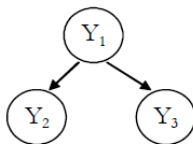


Adding C blocks the fork on $D \rightarrow E \leftarrow C \rightarrow F$

Conditioning set (observed)
 First set
 Second set
 Active (d-connecting) path

NAIVE BAYESIAN CLASSIFIER

- ▶ One simple application of the Bayes network (BN) is the Bayesian classification problem.
- ▶ Consider the DAG



where Y_1 is a categorical variable of mutually exclusive C_1 classes and Y_2 and Y_3 are the two attributes of the objects to be classified. Note that the model assumes that the two attributes are independent given the class $\Rightarrow$ *naive Bayes*.

- ▶ From the above, we have

$$p(y_1, y_2, y_3) = \prod_{i=1}^3 p(y_i | \Pi_i) = p(y_1) p(y_2 | y_1) p(y_3 | y_1).$$

WHAT IS BAYESIAN ABOUT BAYES NETWORKS ?

- ▶ In reality, we are interested in $p(y_1|y_2, y_3)$.
- ▶ One of the reasons, the DAGs are referred to as BNs is their ability to compute different conditional probabilities of interest such as $p(y_1|y_2, y_3)$ in the above.

$$p(y_1|y_2, y_3) \propto p(y_1)p(y_2, y_3|y_1)$$

$$p(y_1|y_2, y_3) \propto p(y_1)p(y_2|y_1)p(y_3|y_1).$$

Note that

$$p(y_1|y_2) \propto p(y_1)p(y_2|y_1)$$

and thus

$$p(y_1|y_2, y_3) \propto p(y_1|y_2)p(y_3|y_1)$$

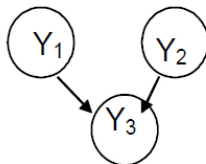
- ▶ The above implies that if you observe y_2 , then you first update $p(y_1)$ to $p(y_1|y_2)$ and this becomes your prior before observing y_3 . Then to obtain $p(y_1|y_2, y_3)$, we use

$$p(y_1|y_2, y_3) \propto p(y_1|y_2)p(y_3|y_1, y_2) = p(y_1|y_2)p(y_3|y_1)$$

$\Rightarrow$ we process observation incoming from one attribute node at a time and this can be generalized to any number of attributes under this structure.

LEARNING IN BAYESIAN NETWORKS

- ▶ Another reason the DAGs are referred to as BNs is due to the fact that we can learn about the unknown parameters of the distributions associated with the node variables.
- ▶ This can be done if we observe data from a specific DAG.
- ▶ There is a third reason: "Bayesian learning about the model structure." First, we focus on a fixed DAG model M and learning about parameters based on data.
- ▶ We are given n observations, that is, n cases from our DAG:



- ▶ We have data $D = \{\mathbf{y}^d = (y_1^d, y_2^d, \dots, y_l^d); d = 1, \dots, n\}$.

MORE NOTATION

- Define $\theta_{ijk} = p(y_{ik}|\pi_{ij}, \theta_{ij})$ = Probability that i th variable is at state k when parent configuration is π_{ij} where $j = 1, \dots, Q_i, k = 1, \dots, C_i$ and

$$\theta_{ij} = (\theta_{ij1}, \dots, \theta_{ijC_i}).$$

is the probability vector for the configuration π_{ij} .

- Consider our example where all three variables are binary.
- Y_1 has no parents so we define $\theta_{1j} = \theta_1 = (\theta_{111}, 1 - \theta_{111})$, where $\theta_{111} = p(y_{11}|\theta_1)$.
- Similarly for Y_2 , we have $\theta_{2j} = \theta_2 = (\theta_{211}, 1 - \theta_{211})$, where $\theta_{211} = p(y_{21}|\theta_2)$.
- For Y_3 , we have 4 different configurations given by:

$$\pi_3 = \{\pi_{31} = (y_{11}, y_{21}), \pi_{32} = (y_{11}, y_{22}), \pi_{33} = (y_{12}, y_{21}), \pi_{34} = (y_{12}, y_{22})\}$$

and $\theta_{31} = (\theta_{311}, 1 - \theta_{311})$, $\theta_{32} = (\theta_{321}, 1 - \theta_{321})$, $\theta_{33} = (\theta_{331}, 1 - \theta_{331})$ and $\theta_{34} = (\theta_{341}, 1 - \theta_{341})$, where $\theta_{3j1} = p(y_{31}|\pi_{3j}, \theta_{3j})$.

IMPLICATIONS

- ▶ At Y_3 , given the parent configuration π_{3j} , we have the conditional probabilities θ_{3j} for $j = 1, \dots, 4$.
- ▶ What are the parent configuration probabilities ?
- ▶ For example, $\pi_{31} = (y_{11}, y_{21})$ and we can evaluate this conditional on θ_1 and θ_2 , that is,

$$p(\pi_{31}|\theta_1, \theta_2) = p(y_{11}, y_{21}|\theta_1, \theta_2).$$

- ▶ Since Y_1 and Y_2 are independent, that is, they are conditionally independent given θ_1, θ_2 , we can write

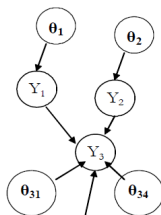
$$p(\pi_{31}|\theta_1, \theta_2) = p(y_{11}, y_{21}|\theta_1, \theta_2) = \theta_{111}\theta_{211}.$$

- ▶ Similarly,

$$p(\pi_{32}|\theta_1, \theta_2) = p(y_{11}, y_{22}|\theta_1, \theta_2) = \theta_{111}(1 - \theta_{211})$$

- ▶ Thus, we really change our DAG by including parameter nodes.

REVISED DAG FOR THE EXAMPLE



- ▶ Note that in the above given θ_1 and θ_2 , we have Y_1 and Y_2 as independent Bernoulli random variables and for each combination of Y_1 and Y_2 , we have a Bernoulli random variable Y_3 given the corresponding θ_{3j} for $j = 1, \dots, 4$.
- ▶ We have also assumed that $\theta_1, \theta_2, \theta_{31}, \theta_{32}, \theta_{33}$ and θ_{34} are mutually independent. This is not necessary but it will makes the estimation easier.
- ▶ The above setup can be generalized for multi-category variables, that is when $C_i > 2$.

LIKELIHOOD FUNCTION

- ▶ Given that we have n cases from the network, that is, $D = \{\mathbf{y}^d; d = 1, \dots, n\}$, we can write down the likelihood function of parameters $\Theta = (\theta_1, \theta_2, \theta_3)$, where $\theta_3 = (\theta_{31}, \theta_{32}, \theta_{33}, \theta_{34})$.
- ▶ In writing the likelihood function $L(\Theta; D)$, we will use the fact that $\mathbf{y}^d$'s are random vectors given Θ and we will use the model structure to write down $p(\mathbf{y}^d|\Theta)$, that is,

$$\begin{aligned} L(\Theta; D) &= \prod_{d=1}^n p(\mathbf{y}^d|\Theta) = \prod_{d=1}^n \prod_{i=1}^3 p(y_i^d|\Pi_i, \Theta) \\ &= \prod_{d=1}^n p(y_1^d|\theta_1) p(y_2^d|\theta_2) p(y_3^d|\pi_3, \theta_3) \end{aligned}$$

Note that since Y_i 's are independent given Π_i and Θ , we can write the above as

$$L(\Theta; D) = \prod_{d=1}^n p(\mathbf{y}^d|\Theta) = \prod_{i=1}^3 \prod_{d=1}^n p(y_i^d|\Pi_i, \Theta)$$

or

$$L(\Theta; D) = \prod_{i=1}^3 L_i(\theta_i; D),$$

and

$$L_i(\theta_i; D) = \prod_{d=1}^n p(y_i^d|\Pi_i, \Theta).$$

LOCALLY INDEPENDENT ESTIMATION

- ▶ The structure of the DAG enables us to decompose the problem to independent estimation problems. In the general setup, this yields independent multinomial likelihoods:

$$L(\Theta; D) = \prod_{i=1}^I L_i(\theta_i; D),$$

where

$$L_i(\theta_i; D) = \prod_{j=1}^{Q_i} \prod_{k=1}^{C_i} (\theta_{ijk})^{N_{ijk}}$$

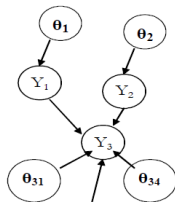
- ▶ In the above, $\theta_{ijk} = p(y_{ik} | \pi_{ij}, \theta_{ij})$ and we define N_{ijk} = Number of cases in which variable i takes the value y_{ik} and the parents of the variable are in configuration π_{ij} . The MLEs of θ_{ijk} 's can be obtained as

$$\widehat{\theta}_{ijk} = \frac{N_{ijk}}{N_{ij}},$$

where $N_{ij} = \sum_{k=1}^{C_i} N_{ijk}$, that is, number of cases where parent configuration is π_{ij} .

DIRICHLET PRIORS

- In the example DAG



we assumed that elements of $\Theta = (\theta_1, \theta_2, \theta_{31}, \theta_{32}, \theta_{33}, \theta_{34})$ are independent.

- If we make the same assumption in the general case and assume independent Dirichlet priors for the elements of Θ , we have a conjugate Bayesian analysis. More specifically, for each parent configuration π_{ij} for variable i , we assume that parameter vector $\theta_{ij} = (\theta_{ij1}, \dots, \theta_{ijC_i})$ has a Dirichlet distribution with parameters $(\alpha_{ij1}, \dots, \alpha_{ijC_i})$,

$$p(\theta_{ij}) \propto \prod_{k=1}^{C_i} (\theta_{ijk})^{\alpha_{ijk}-1}.$$

POSTERIOR ANALYSIS

- Thus, for all Q_i configurations, we have

$$p(\theta_i) \propto \prod_{j=1}^{Q_i} \prod_{k=1}^{C_i} (\theta_{ijk})^{\alpha_{ijk}-1}$$

and the prior for Θ is

$$p(\Theta) \propto \prod_{i=1}^I \prod_{j=1}^{Q_i} \prod_{k=1}^{C_i} (\theta_{ijk})^{\alpha_{ijk}-1} = \prod_{i=1}^I \prod_{j=1}^{Q_i} p(\theta_{ij})$$

- This results in "local posterior estimation", that is,

$$p(\Theta \mid D) = \prod_{i=1}^I \prod_{j=1}^{Q_i} p(\theta_{ij} \mid D),$$

where

$$p(\theta_{ij} \mid D) \propto \prod_{k=1}^{C_i} (\theta_{ijk})^{\alpha_{ijk}-1} (\theta_{ijk})^{N_{ijk}}.$$

is a Dirichlet with parameters $(\alpha_{ij1} + N_{ij1}, \dots, \alpha_{ijC_i} + N_{ijC_i})$

SELECTION OF PRIOR PARAMETERS

- Note that in specifying the prior parameters of the Dirichlet model

$$p(\theta_{ij}) \propto \prod_{k=1}^{C_i} (\theta_{ijk})^{\alpha_{ijk}-1}$$

small values of α_{ijk} represents greater uncertainty. Thus, a local prior precision parameter is given by

$$\alpha_{ij} = \sum_{k=1}^{C_i} \alpha_{ijk}$$

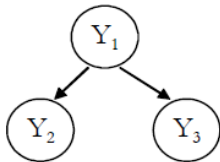
- A common (global) precision parameter α (the *equivalent sample size*) is assigned to every node, that is,

$$\sum_{j=1}^{Q_i} \alpha_{ij} = \alpha \quad \text{for all } i = 1, \dots, I.$$

"Initial ignorance" can be represented by choosing $\alpha_{ijk} = \frac{\alpha}{C_i Q_i}$, which implies $\alpha_{ij} = \alpha / Q_i$ and $E(\theta_{ijk}) = \alpha_{ijk} / \alpha_{ij} = 1 / C_i \Rightarrow$ "uniform type prior".

EXAMPLE: BRITISH ELECTION SURVEY 1992

- ▶ Consider a classification problem



- ▶ Voting intention (Y_1 : 1 = Conservative, 2 = Labour, 3 = Liberal Democrat, 4 = Other)
- ▶ Gender (Y_2 : 1=male, 2=female)
- ▶ Social Class (Y_3 : 1=low, 2=middle, and 3 = high)
- ▶ Goal: To classify the voters into four classes of voting intentions given the two attributes.
- ▶ We have data presented in contingency table format.

SURVEY DATA (Ramoni&Sebastiani 1998)

		Y_1			
Y_2	Y_3	1	2	3	4
1	1	28	8	7	0
	2	153	114	53	14
	3	20	31	17	1
2	1	1	1	0	1
	2	165	86	54	6
	3	30	57	18	4

- Define

$$\theta_1 = (\theta_{111}, \theta_{112}, \theta_{113}, \theta_{114})$$

$$\theta_{2j} = (\theta_{2j1}, \theta_{2j2}), j = 1, \dots, 4 \text{ (4 configurations of } Y_1)$$

$$\theta_{3j} = (\theta_{3j1}, \theta_{3j2}, \theta_{3j3}), j = 1, \dots, 4 \text{ (4 configurations of } Y_1)$$

- Let's choose global precision $\alpha = 12$ and use $\alpha_{ijk} = \frac{\alpha}{C_i Q_i}$
 $\Rightarrow \alpha_{1jk} = (3, 3, 3, 3), \alpha_{2jk} = (1.5, 1.5)$ and $\alpha_{3jk} = (1, 1, 1)$.

POSTERIOR UPDATING

$$\theta_1|D \sim \text{Dir}(3 + 397, 3 + 297, 3 + 149, 3 + 26)$$

$$\theta_{21}|D \sim \text{Dir}(1.5 + 28 + 153 + 20, 1.5 + 196)$$

$$\theta_{22}|D \sim \text{Dir}(1.5 + 153, 1.5 + 144)$$

$$\theta_{23}|D \sim \text{Dir}(78.5, 73.5)$$

$$\theta_{24}|D \sim \text{Dir}(16.5, 12.5)$$

$$\theta_{31}|D \sim \text{Dir}(1 + 28 + 1, 1 + 318, 1 + 50)$$

$$\theta_{24}|D \sim \text{Dir}(1 + 9, 1 + 200, 1 + 88)$$

$$\theta_{24}|D \sim \text{Dir}(1 + 7, 1 + 107, 1 + 35)$$

$$\theta_{24}|D \sim \text{Dir}(1 + 1, 1 + 20, 1 + 5)$$

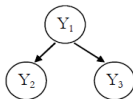
- What will be the posterior classification probabilities ?

$$p(y_1|y_2, y_3, D) \propto p(y_1|D)p(y_2|y_1, D)p(y_3|y_1, D),$$

where these can be computed using the expected posterior means of the Dirichlet distribution.

POSTERIOR PROBABILITIES

Y_1			
1	2	3	4
0.454	0.340	0.173	0.033



Y_2		
Y_1	1	2
1	0.506	0.494
2	0.515	0.485
3	0.516	0.484
4	0.569	0.431

Y_3			
Y_1	1	2	3
1	0.075	0.798	0.127
2	0.033	0.670	0.297
3	0.053	0.710	0.237
4	0.069	0.724	0.207

- Now we can obtain classification probabilities using terms such as

$$p(y_1 | Y_2 = 1, Y_3 = 1, D) \propto p(y_1 | D) p(Y_2 = 1 | y_1, D) p(Y_3 = 1 | y_1, D)$$

POSTERIOR CLASSIFICATION PROBABILITIES

- ▶ We can obtain this for

$$p(Y_1 = j | Y_2 = 1, Y_3 = 1, D) \propto p(Y_1 = j | D) p(Y_2 = 1 | Y_1 = j, D) p(Y_3 = 1 | Y_1 = j, D)$$

for $j = 1, \dots, 4$ and then normalize, that is,

$$p(Y_1 = 1 | Y_2 = 1, Y_3 = 1, D) \propto 0.454 \times 0.506 \times 0.075 = \mathbf{0.593}$$

$$p(Y_1 = 2 | Y_2 = 1, Y_3 = 1, D) \propto 0.340 \times 0.515 \times 0.033 = \mathbf{0.199}$$

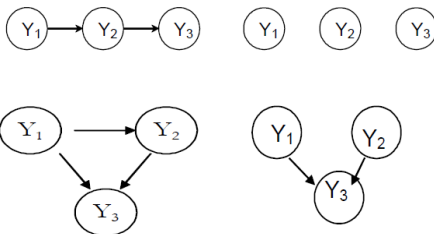
$$p(Y_1 = 3 | Y_2 = 1, Y_3 = 1, D) \propto 0.173 \times 0.516 \times 0.053 = \mathbf{0.163}$$

$$p(Y_1 = 4 | Y_2 = 1, Y_3 = 1, D) \propto 0.033 \times 0.569 \times 0.069 = \mathbf{0.045}$$

- ▶ The above (numbers in bold) are the normalized probabilities.

LEARNING ABOUT THE MODEL FROM DATA

- ▶ Another use of Bayes nets is for learning about the model structure from data
⇒ learning about "causal relationships".
- ▶ This is very similar to ideas in Bayesian model selection.
- ▶ How does a Bayesian select a model ?
 - ▶ Given a set of nodes $Y = \{Y_1, \dots, Y_I\}$, every DAG is a candidate model. If we have three nodes, then DAGs



are possible models (and there are more).

MARGINAL LIKELIHOODS

- ▶ If we have G candidate models, $M_1, \dots, M_G$ and have prior model probabilities as $p(M = M_g) = p(M_g)$, then posterior model probabilities can be obtained in the usual manner as

$$p(M_g|D) \propto p(D|M_g)p(M_g)$$

- ▶ Note that $p(D|M_g)$ is the marginal likelihood term for model M_g which can be obtained as

$$p(D|M_g) = \int p(D|M_g, \Theta)p(\Theta|M_g)d\Theta$$

- ▶ Under the Dirichlet prior and local and global parameter independence, the above term can be analytically obtained for any model. Given its parents, the result for any node Y_i is a Dirichlet-multinomial result we had earlier

$$p(D_i|M_g) = \prod_{j=1}^{Q_i} \frac{\Gamma(\alpha_{ij})}{\Gamma(\alpha_{ij} + N_{ij})} \prod_{k=1}^{C_i} \frac{\Gamma(\alpha_{ijk} + N_{ijk})}{\Gamma(\alpha_{ijk})}$$

and

$$p(D|M_g) = \prod_{i=1}^I p(D_i|M_g)$$

COMPUTATION OF MARGINAL LIKELIHOOD

- ▶ Note that due to its decomposability computation of $p(D|M_g)$ is straightforward but unfortunately as the number of nodes increases it becomes infeasible to evaluate all possible models $\Rightarrow$ need "intelligent algorithms".
- ▶ Search algorithms try to exploit decomposability of $p(D|M_g)$.
- ▶ Basic idea: For each node Y_i start adding a parent at a time and compute the local likelihood contribution $L(Y_i; \Pi_i) = p(D_i|M_g)$. The parent set Π_i can be expanded by including the parent that gives the largest contribution to $L(Y_i; \Pi_i)$.
- ▶ In some cases, there may be prior information about the structure and this can also be used in the algorithm.
- ▶ For example, we may know that Y_i can not be a parent of Y_j .
- ▶ In a three node problem $Y = \{Y_1, Y_2, Y_3\}$, we may know that Y_3 cannot be a parent of Y_2 and Y_2 cannot be a parent of Y_1 .

A MODEL SEARCH EXAMPLE

- Consider the following example

		Y_3	
Y_1	Y_2	1	2
1	1	2	10
	2	5	3
2	1	6	3
	2	10	2

- Note that all the variables are binary. Obviously, we can compare few candidate models by computing their marginal likelihoods.
- Instead, assume that we know Y_3 cannot be a parent of Y_2 and Y_2 cannot be a parent of Y_1 , that is, Y_3 can only be a child and specify global precision as $\alpha = 6$.
- Let's start with Y_3 and compute $L(Y_3; \Pi_3)$ for $\Pi_3 = \emptyset$, $\Pi_3 = \{Y_1\}$ and $\Pi_3 = \{Y_2\}$.

EXAMPLE DETAILS

- ▶ When $\Pi_3 = \emptyset$

$$L(Y_3; \Pi_3) = \frac{\Gamma(\alpha_{31})}{\Gamma(\alpha_{31} + N_{31})} \prod_{k=1}^2 \frac{\Gamma(\alpha_{31k} + N_{31k})}{\Gamma(\alpha_{31k})}$$

$$\alpha_{311} = \alpha_{312} = 6/2 = 3 \Rightarrow \alpha_{31} = 3 + 3 = 6.$$

$$N_{311} = 23 \text{ and } N_{312} = 18 \Rightarrow N_{31} = 41.$$

$$L(Y_3; \Pi_3) = \frac{\Gamma(6)}{\Gamma(47)} \frac{\Gamma(26)}{\Gamma(3)} \frac{\Gamma(21)}{\Gamma(3)} \Rightarrow \ln(L(Y_3; \Pi_3)) = -29.21$$

- ▶ When $\Pi_3 = \{Y_1\}$

$$L(Y_3; \Pi_3) = \prod_{j=1}^2 \frac{\Gamma(\alpha_{3j})}{\Gamma(\alpha_{3j} + N_{3j})} \prod_{k=1}^2 \frac{\Gamma(\alpha_{3jk} + N_{3jk})}{\Gamma(\alpha_{3jk})}.$$

$$\alpha_{311} = \alpha_{312} = \alpha_{321} = \alpha_{322} = 6/4 = 1.5 \Rightarrow$$

$$\alpha_{31} = \alpha_{32} = 1.5 + 1.5 = 3$$

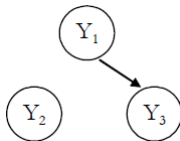
$$Y_1 = 1: N_{311} = 7 \text{ and } N_{312} = 13 \Rightarrow N_{31} = 20.$$

$$Y_1 = 2: N_{321} = 16 \text{ and } N_{322} = 5 \Rightarrow N_{32} = 21.$$

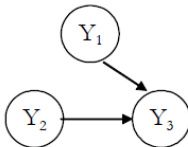
We can show that $\ln(L(Y_3; \Pi_3)) = -27.05$.

- ▶ One can compute all these terms using Bayes network packages such as R-package *bnlearn*.

- ▶ We can also compute it for $\Pi_3 = \{Y_2\}$ which gives here a slightly lower value as $Ln(L(Y_3; \Pi_3)) = -27.72$ and therefore, we expand from $\Pi_3 = \emptyset$ to $\Pi_3 = \{Y_1\}$.
- ▶ Now we have the diagram revised as



- ▶ Next the algorithm checks the above, against $\Pi_3 = \{Y_1, Y_2\}$, that is, against



- ▶ The above can be shown as the best model with the highest score of $Ln(L(Y_3; \Pi_3)) = -26.81$ after also checking against $\Pi_2 = \{Y_1\}$ which is not supported.